Example: Suppose That the temperature at a point (x, y, z) in space is given by the following relationship: $T(x,y,z) = \frac{80}{1+x^2+2y^2+3z^2}$, where T is measured in degrees celsius and x, y, z in meters. In which direction does the temperature increase fastest at The point $(1,1,-2)$? what is the Max rate of increase?

$$\nabla T = \left\langle \frac{-80(2x)}{(1+x^2+2y^2+3z^2)^2}, \frac{-80(4y)}{(1+x^2+2y^2+3z^2)^2}, \frac{-80(6z)}{(1+x^2+2y^2+3z^2)^2} \right\rangle$$

$$\Rightarrow \nabla T = \frac{160}{(1+x^2+2y^2+3z^2)^2} \langle -x, -2y, -3z \rangle, \quad \nabla T(1,1,-2) = \frac{160}{(1+1^2+2(1)^2+3(-2)^2)^2} \langle -1, -2, 6 \rangle$$

$$\Rightarrow \nabla T(1,1,-2) = \frac{5}{8}(-i - 2j + 6k)$$

The temperature increases fastest in this direction

$$\text{The Max rate} = |\nabla T| = \frac{5}{8}\sqrt{1+4+36} = \frac{5\sqrt{41}}{8} \approx 4^\circ C/m$$

Q :- A drone starts at pt $(1,2,3)$ & moves in the direction of the vector $\vec{v} = \langle 2, -1, 2 \rangle$ with a constant speed of 5 units/sec.

(a) Find position vector $\vec{v}(t)$ of the drone at time 't'.

(b) Find position after 4 sec.

Sol

$$\vec{r}_0 = \langle 1, 2, 3 \rangle$$

$$|\vec{v}| = \sqrt{(2)^2 + (-1)^2 + (2)^2} = \sqrt{9} = 3$$

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \left\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right\rangle$$

$$\vec{r}(t) = \langle 1, 2, 3 \rangle + t(5)\left\langle \frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right\rangle$$

$$\vec{r}(t) = \langle 1, 2, 3 \rangle + t\left\langle \frac{10}{3}, -\frac{5}{3}, \frac{10}{3} \right\rangle$$

$$\vec{r}(4) = \langle 1, 2, 3 \rangle + 4\left\langle \frac{10}{3}, -\frac{5}{3}, \frac{10}{3} \right\rangle$$

$$\vec{r}(4) = \left\langle \frac{43}{3}, \frac{-14}{3}, \frac{49}{3} \right\rangle$$

Q:- A security laser beam travels along the line

$x(t) = 1+2t, \quad y(t) = 1+t, \quad z(t) = 2t$

A hacker tries to cross at pt $S = (3,2,1)$.

a) Find the shortest distance b/w hacker & laser beam.

(b) Will the hacker trigger the alarm if detection radius is 1 unit?

Sol If $t=0$ $P=(1,1,0)$, $\vec{V} = (2,1,2)$

$$\vec{PS} = \langle 2,1,1 \rangle$$

$$\vec{PS} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 2 & 1 & 2 \end{vmatrix}$$

$$= \hat{i}(2-1) - j(4-2) + k(2-2)$$

$$= \hat{i} - 2\hat{j}$$

$$|\vec{PS} \times \vec{v}| = \sqrt{1+4} = \sqrt{5} \qquad |\vec{v}| = \sqrt{4+1+4} = 3$$

$$d = \frac{|\vec{PS} \times \vec{v}|}{|\vec{v}|}$$

$$d = \frac{\sqrt{5}}{3}$$

Since $\frac{\sqrt{5}}{3} < 1$ so alarm will trigger (Distance $\leq$ Detection Radius)

Q:- A security laser beam protecting a runway follows:

$$x = 2+3t, \quad y = 1-t, \quad z = 4+2t$$

An unauthorized drone is detected at:

$$S = (8, -1, 10)$$

The laser detection radius is 1.5 units. Will the alarm trigger.

sol

If $t = 0$ $\quad P = (2, 1, 4)$

$$\vec{v} = \langle 3, -1, 2 \rangle$$

$$\vec{PS} = \langle 6, -2, 6 \rangle$$

$$\vec{PS} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & -2 & 6 \\ 3 & -1 & 2 \end{vmatrix}$$

$$= i(-4+6) - j(12-18) + k(-6+6)$$

$$= 2\hat{i} + 6\hat{j}$$

$$|\vec{PS} \times \vec{v}| = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10}$$

$$|\vec{v}| = \sqrt{9+1+4} = \sqrt{14}$$

$$d = \frac{2\sqrt{10}}{\sqrt{14}}$$

$$d \approx 1.69$$

Since $1.69 > 1.5$

Alarm will not trigger.

Q:- The temp T at a pt (x,y,z) in space is inversely proportional to the square of the distance from (x,y,z) to the origin. It is known that $T(0,0,1) = 500$. Find rate of change of T at $(2,3,3)$ in the direction of $(3,1,1)$. In which direction from $(2,3,3)$ does the temp T increase most rapidly? At $(2,3,3)$ what is the max rate of change of T?

Sol Distance from origin : $r = \sqrt{x^2+y^2+z^2}$

$$T \propto \frac{1}{r^2}$$

$$T = \frac{k}{x^2+y^2+z^2}$$

$$T(0,0,1) = 500$$

$$500 = \frac{k}{1^2}$$

$$\boxed{k = 500}$$

$$T(x,y,z) = \frac{500}{x^2+y^2+z^2}$$

$$\vec{\nabla}T = -500(x^2+y^2+z^2)^{-2}(2x,2y,2z)$$

$$\vec{\nabla}T = \frac{-1000(x,y,z)}{(x^2+y^2+z^2)^2}$$

grad at $(2,3,3) \Rightarrow x^2+y^2+z^2 = 4+9+9 = 22$

$$\vec{\nabla}T(2,3,3) = -\frac{1000}{(22)^2}(2,3,3)$$

$$= \left(\frac{-2000}{484}, \frac{-3000}{484}, \frac{-3000}{484}\right)$$

Rate of change in the direction of $(3,1,1)$

$$\vec{U} = \frac{1}{\sqrt{11}}(3,1,1)$$

$$D_{\vec{U}}T = \vec{\nabla}T \cdot \vec{U}$$

$$= \frac{1}{\sqrt{11}}\left[\frac{-2000}{484}(3) + \frac{(-3000)}{484}(1) + \frac{(-3000)}{484}(1)\right]$$

$$= \frac{-12000}{484\sqrt{11}}$$

Temp decreases in this direction

Temp increases most rapidly in the direction of gradient

The direction vector is $\vec{U} = \langle -2,-3,-3 \rangle$

$$\vec{U} = \frac{1}{\sqrt{22}}(-2,-3,-3)$$

Temp increases fastest towards the origin.

Max rate of change $= |\vec{\nabla}T| = \frac{1000}{(22)^2}\sqrt{(2)^2+(3)^2+(3)^2}$

$$|\vec{\nabla}T(2,3,3)| = \frac{1000\sqrt{22}}{484}$$

Q:- The temp at a pt (x,y,z) in a metal solid is

$$T(x,y,z) = \frac{xyz}{1+x^2+y^2+z^2}$$

i) Find rate of change of temp wrt distance at $(1,1,1)$ in the direction of origin.

Sol

$f = xyz \quad , \quad g = 1+x^2+y^2+z^2$

$\nabla T = \frac{g\nabla f - f\nabla g}{}$

At $(1,1,1)$

$f = 1$

$g = 4$

$\nabla f = (yz, xz, xy) = (1,1,1)$

$\nabla g = (2,2,2)$

$$\nabla T = \frac{4(1,1,1) - 1(2,2,2)}{16} = \frac{(2,2,2)}{16} = \left\langle \frac{1}{8}, \frac{1}{8}, \frac{1}{8} \right\rangle$$

Direction from $(1,1,1)$ to origin

$$\vec{U} = \frac{1}{\sqrt{3}}(-1,-1,-1)$$

$$D_{\vec{U}} T = \vec{\nabla} T \cdot \vec{U}$$

$$= \frac{-3}{8\sqrt{3}}$$

Temp decreases at rate of $\frac{3}{8\sqrt{3}}$

(b) Find direction in which temp rises most rapidly at pt. $(1,1,1)$. Express answer as unit vector.

Sol Temp inc in the direction of gradient

$$\vec{\nabla T}(1,1,1) = \left(\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right)$$

Unit vector $= \vec{U} = \frac{1}{\sqrt{3}}(1,1,1)$

(c) Find rate at which temp rises moving from $(1,1,1)$ in the direction obtained in (b).

Sol

$$|\vec{\nabla} T(1,1,1)| = \frac{\sqrt{3}}{8}$$